Limnology and Oceanography

Supporting Information for:

Future inorganic nutrient and plankton dynamics in a temperate lake

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S1. Model configuration and equations

Physical configuration

The lake ecosystem model operates in a zero-dimensional physical configuration that consists of two layers stacked on top of each other, representing the upper epilimnion and the bottom hypolimnion (Fig. 1) and reflecting the so-called slab physics approximation (e.g. Fasham et al. 1990; To et al. 2024),. The depth of the epilimnion is known as the mixed layer depth (MLD) and varies seasonally. The epilimnion is biologically active and is assumed to be well-mixed so that all model properties, physical and biological, are considered homogeneously distributed within its volume. The hypolimnion does not include explicit plankton dynamics. In the hypolimnion, a fixed amount of nutrient, P_0, representing a constant reservoir of inorganic nutrient driven by unspecified recycling processes, replenishes the inorganic phosphorus nutrient pool, P, of the epilimnion through physical mixing. Changes in motile (zooplankton) and non-motile (nutrient, detritus, and phytoplankton) model variables, induced by physical processes, can occur due to (1) exchange of fluxes driven by diffusive mixing between the epilimnion and hypolimnion and (2) variations in the MLD. The variation in zooplankton biomass due to changes in the MLD is described by the following term (Evans and Parslow 1985):

$$\lambda_z = \frac{h}{MLD(t)}, \tag{S1}$$

with $h = \frac{dMLD(t)}{dt}$. Here we assume that zooplankton can maintain their position within the upper epilimnion. The shallowing or deepening of the MLD will, respectively, increase or decrease zooplankton concentrations. A shallowing MLD leaves the concentrations of non-motile variables (i.e. nutrient, phytoplankton, and detritus) unchanged because their reduction in number due to detrainment is compensated by the reduction in volume of the epilimnion. A deepening MLD reduces the concentrations of non-motile variables, unless P_0 is larger than P, that is, when the nutrient input from the hypolimnion is greater than the nutrient concentration in the epilimnion. The non-motile variables are also subject to diffusive mixing. The variations in biomasses of non-motile variables due to physical processes can be written as follows (Evans and Parslow 1985):

$$\lambda = \frac{\omega + h^+(t)}{MLD(t)}, \tag{S2}$$

with ω being the diffusive mixing constant and

$$h^+(t) = \max(0, h). \tag{S3}$$

Biological components

The model consists of a set of 14 ordinary differential equations:

$$\frac{dP}{dt} = -\sum_i \mu_i A_i + (1-\gamma)\varepsilon \sum_i \sum_j G_{ij} Z_j + \varphi D + \lambda(P_0 - P) \quad \text{(S4)}$$

$$\frac{dA_i}{dt} = (\mu_i - \phi_A - \lambda)A_i - \sum_i \sum_j G_{ij} Z_j \quad \text{(S5)}$$

$$\frac{dZ_j}{dt} = \left(\varepsilon\gamma \sum_i \sum_j G_{ij} Z_j - \phi_Z - \lambda_z - \eta_Z Z_j \right) Z_j \quad \text{(S6)}$$

$$\frac{dD}{dt} = \sum_i \phi_P A_i + \sum_j \phi_z Z_j + (1-\varepsilon) \sum_i \sum_j G_{ij} Z_j - (\varphi + \lambda) D \quad \text{(S7)}$$

with $i = 1, \dots, 10$ and $j = 1, 2$. All the terms in the equations are described in the following.

The dynamics of phosphorus (Eq. S4) is driven by nutrient lost to phytoplankton growth, $\sum_i \mu_i A_i$, and nutrient replenishment from (1) the portion of grazed materials that is not assimilated by zooplankton, $(1-\gamma)\varepsilon \sum_i \sum_j G_{ij} Z_j$, (2) the remineralization of detritus, φD, and (3) the exchange between the epilimnion and hypolimnion via vertical mixing, $\lambda(P_0 - P)$, where P_0 is the nutrient concentration in the hypolimnion. We determined the value P_0 based on average winter observations, when the entire water column is well mixed and nutrient concentrations are homogeneously distributed throughout it.

The temporal dynamics of the ith phytoplankton size class (Eq. S5) is regulated by phytoplankton growth, $\mu_i A_i$, and by several loss processes. The growth rate of phytoplankton, described by,

$$\mu_i = \mu_{max_i}(S_i^A) \cdot E(T) \cdot U(P) \cdot H(I_z)\,, \quad \text{(S8)}$$

is regulated by a size-dependent maximum growth rate term,

$$\mu_{max_i}(S_i^A) = \beta_{\mu_{max}} \cdot \left(\frac{S_i^A}{1\ \mu\text{m}} \right)^{\alpha_{\mu_{max}}}, \quad \text{(S9)}$$

with S_i^A representing the size of the ith phytoplankton size class (Fig. 2A), and three environmental factors: temperature T, phosphorus P, and light I_z. The temperature-regulated growth term, $E(T)$, following Norberg (2004), a new feature considered in this model, is presented in the main article, section 'Temperature dependences'. The nutrient uptake term follows a Monod function (Monod 1949),

$$U(P) = \frac{P}{K_P + P}, \tag{S10}$$

with P representing the inorganic nutrient concentration in the lake (Eq. S4) and K_P the half-saturation constant for nutrient.

The light-limited growth term depends on the daily MLD-averaged photosynthetic rate,

$$H(I_z) = \frac{1}{MLD}\int_{z=0}^{z=MLD} P(I_z)dz\,, \tag{S11}$$

and is calculated by integrating the photosynthesis-irradiance curve (Lewis and Smith, 1983),

$$P(I_z) = P_{max}\frac{\alpha_{P_I}\cdot I_z}{\sqrt{{P_{max}}^2 + \left(\alpha_{P_I} I_z\right)^2}}, \tag{S11a}$$

through depth z (Eq. 11a). In this equation, α_{P_I} is the initial slope of the curve, P_{max} is the maximum photosynthetic rate, and I_z is a function estimating the level of irradiance, I, at depth z, based on the Beer-Lambert's law:

$$I_z = I_0 \cdot e^{-K_{par}\cdot z}. \tag{S11b}$$

K_{par} is the light attenuation coefficient due to water, and I_0 is the irradiance value at surface, i.e., the observed PAR prescribed as forcing to the model. To optimize numerical performance, $H(I_z)$ is computed based on an analytical approximation proposed by Anderson et al. (2015):

$$H(I_z) = \frac{P_{max}}{K_{par}MLD}\log\left(\frac{\alpha_{P_I}I_0 + \sqrt{{P_{max}}^2 + \left(\alpha_{P_I}I_0\right)^2}}{\alpha_{P_I}I_z + \sqrt{{P_{max}}^2 + \left(\alpha_{P_I}I_z\right)^2}}\right). \tag{S11c}$$

Phytoplankton is lost due to natural mortality, ϕ_A, mixing, λ, and zooplankton grazing $\sum_i \sum_j G_{ij}\, Z_j$.

The temporal dynamics for the jth zooplankton size class (Eq. S6) is regulated by net grazing, $\varepsilon\gamma \sum_i \sum_j G_{ij}\, Z_j$, which is the portion of grazed phytoplankton successfully ingested and assimilated, and several loss processes. The grazing rate of zooplankton is determined by,

$$G_{ij} = \psi_{max_j}\left(S_j^Z\right) \cdot \psi(T) \cdot \left(\frac{\delta_{ij}\left(S_i^A,\, S_j^Z\right) A_i}{K_A + \sum_i \delta_{ij}\left(S_i^A,\, S_j^Z\right) A_i}\right). \tag{S12}$$

The first factor describes the size-dependent maximum ingestion rate:

$$\psi_{max_j}\left(S_j^Z\right) = \beta_{\psi_{max}} \cdot {S_j^Z}^{\alpha_{\psi_{max}}}, \tag{S13}$$

with S_j^Z representing the size of the jth zooplankton size class (Fig. 2C). The second factor, $\psi(T)$, describes the temperature dependence of grazing and is presented in detail in the main article, section 'Temperature dependences' (Eq. 2). The third factor of the grazing formulation comprises a half-saturation constant for phytoplankton intake, K_A, and the grazing preference for a specific range of phytoplankton size classes,

$$\delta_{ij}\left(S_i^A, S_j^Z\right) = e^{-\left[\frac{log_{10}\left(S_i^A\right) - log_{10}\left(P_{opt_j}\left(S_j^Z\right)\right)}{\theta_j}\right]^2}. \tag{S14}$$

The grazing preference is characterized by a Gaussian curve over a specific range of phytoplankton size classes. The median and the width of this Gaussian are determined by, respectively, the optimal phytoplankton size, $A_{opt_j}\left(S_j^Z\right)$, and the level of selectivity of the grazer θ_j (Fig. 2B). Grazers of different sizes are characterized by different optimal phytoplankton sizes following an allometric scaling as:

$$A_{opt_j}\left(S_j^Z\right) = \beta_{P_{opt}} \cdot {S_j^Z}^{\alpha_{A_{opt}}}. \tag{S15}$$

This function is empirically derived from the data of Hansen et al. (1997), reflecting the fact that large zooplankton prefer large phytoplankton cells (Fig. 2D).

The loss processes for zooplankton include natural mortality, ϕ_Z, mixing, λ_z, and higher order predation, $\eta_Z Z_j$ (Eq. S6). The portion of grazed phytoplankton that is not ingested due to sloppy feeding, $(1-\varepsilon)\sum_i \sum_j G_{ij} Z_j$, flows into the detritus pool (Eq. S7). The detritus pool acts as a nutrient recycling station by remineralizing dead plankton and non-ingested materials from grazing. The portion that is ingested but not assimilated, $(1-\gamma)\varepsilon\sum_i \sum_j G_{ij} Z_j$, is excreted and quickly remineralized into the nutrient pool (Eq. S4).

S2. Environmental forcing

The model is forced with environmental variables observed at Greifensee, a medium-size (about 8.45 km^2, with a maximum depth of 33 m) eutrophic lake near Zurich (Switzerland). The environmental variables are: water temperature, Photosynthetically Active Radiation (PAR), and Mixed Layer Depth (MLD). Water temperature is obtained from the CTD profiles as daily average epilimnetic temperature from March 2019 to December 2022. The PAR data is converted from the solar irradiance data collected daily by MeteoSwiss, from a weather station next to Greifensee, for the same period as the water temperature. We retrieved data for mixed layer depth from Pomati et al. (2020), which documented monthly values at Greifensee from 1972 to 2015. The forcing functions for the model are daily values over a one-year period. We used daily means from a three-year dataset (March 2019 to December 2022) for water temperature, 75-percentiles from the same three-year dataset for PAR, and daily values obtained by interpolating the monthly averages for MLD (Fig. S1). Details of the procedures followed for generating the forcing functions are reported in section S2 of Supporting Information and the environmental data are provided in Supplementary Data.

Data source

These forcing functions are derived from various time-series data, and the data cover different time periods. Water temperatures are obtained from the CTD profiles as daily average epilimnetic temperature from 2019 to 2022, measured by the Swiss Federal Institute of Aquatic Science and Technology, Eawag. The PAR data is converted from solar radiance data collected from a weather station next to Greifensee, measured daily by MeteoSwiss for the same period as the water temperature. The PAR conversion is presented in the next paragraph. Finally, we obtained MLD data from Pomati et al. (2020). These are monthly data collected at Greifensee from 1972 to 2015 (n=519). To match the time step of the model (one day), we obtained daily values by interpolating the monthly averages for MLD (see below).

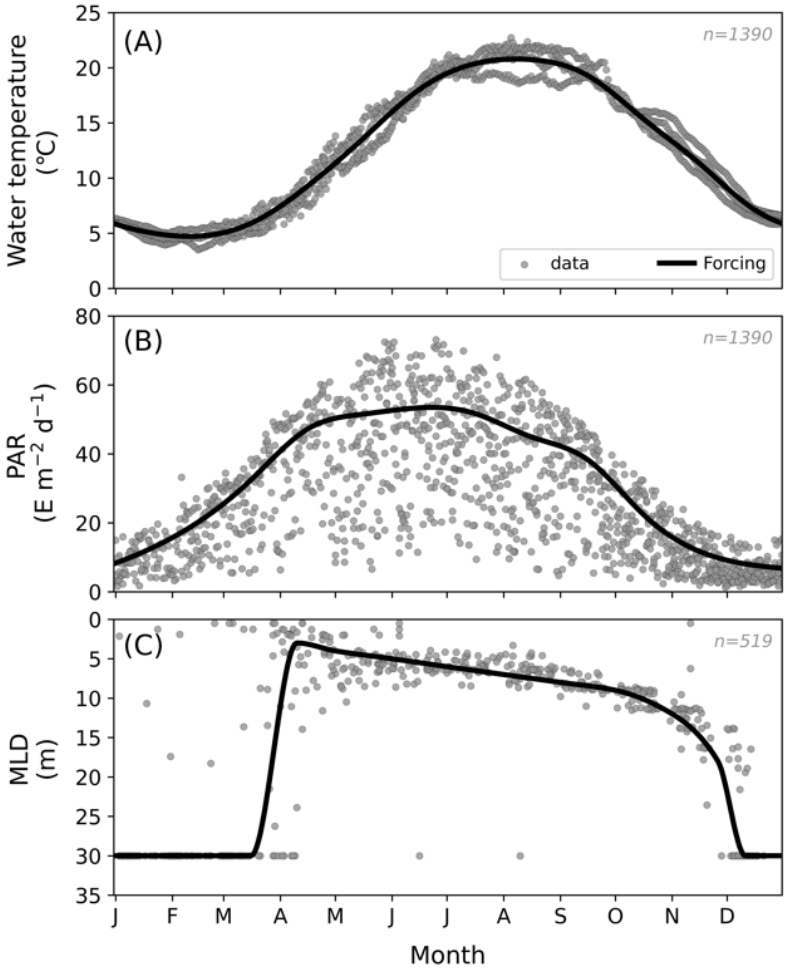
Unit conversion of PAR

PAR values for plankton models are often obtained from light measurements, and light can be expressed as photon flux density (i.e. Ein m^{-2} d^{-1} / mol quanta m^{-2} s^{-1}), or as energy flow density (i.e. W m^{-2}) (Lampert and Sommer, 2007). The photosynthetic processes of modelled phytoplankton are often expressed in Ein m^{-2} d^{-1} or mol quanta m^{-2} s^{-1}, whereas solar irradiance is commonly measured in W m^{-2}. First, we calculate the amount of PAR from the solar irradiance in terms of net short-wave solar irradiance, because not all wavelengths of the light spectrum are usable by photosynthesis. Only wavelengths in the range 0.4-0.7 μm are suitable for most phytoplankton to perform photosynthesis, and PAR is often considered in the range 0.41 to 0.50 of the net short-wave solar irradiance (Howell

et al. 1983; Tsubo and Walker 2005). The specific value depends on climatic conditions and differs geographically. We adopt an intermediate value of 0.45 (Table S2, S3). After obtaining PAR in units of W m^{-2}, we transformed these units into Ein m^{-2} d^{-1}. From Lampert and Sommer (2007), an approximated conversion from W m^{-2} to µEin m^{-2} s^{-1} to ranges from 4 to 5. We adopt the value of 5 (Table S2, S3).

Derivations of forcing function

Based on the monthly values of the three sets of environmental data, we derive corresponding annual forcing functions for water temperature, PAR, and MLD (Fig. S1). The water temperature is obtained from CTD profiles as daily average epilimnetic temperatures from March 2019 to December 2022. After obtaining monthly means from the daily averages over a one-year period, we interpolated the monthly values to derive the forcing function for water temperature. The PAR data exhibits a wide range of variability in the annual composite (Fig. S1), a common feature for irradiance data due to high inter-annual or daily variabilities driven by cloud cover. We obtained the monthly means based on the 75-percentile of the daily averages over a one-year period and interpolated the monthly values to derive the forcing function for PAR (Fig. S1). The MLD data are retrieved from a monthly time-series over 40 years (1972-2015) from Pomati et al. (2020). We manually derived twelve monthly data points that capture the changes in the mixed layer depth and interpolate the monthly data into daily values. The interpolations of the environmental data were performed using piecewise polynomials (i.e. Spline) in the *scipy* package (Python 3.7.x) (Virtanen et al. 2020). The code for generating the function is available on Github (https://github.com/systemsecologygroup/TempSizeMod). The environmental data points and the generated forcing functions used as input to the model are shown in Fig. S1.



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191 **Fig. S1** Forcing functions used in our model. (A) Water temperature (n = 1390). (B) Photosynthetically Active

192 Radiation, PAR (n = 1390). (C) Mixed Layer Depth, MLD (n = 519).

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S3. Model calibration

The model was calibrated by daily mean nutrient concentrations (phosphate) and phytoplankton and zooplankton abundance and size data, provided by the Swiss Federal Institute of Aquatic Science and Technology (Eawag). These data covered the period from March 2019 to December 2022. We selected the six most dominant (in terms of abundances) groups of phytoplankton at Greifensee, namely, chlorophytes, rhodomonas, cryptophyceae, centric diatoms, dinobryon, and uroglena, and two dominant groups of herbivorous zooplankton, namely, ciliates and daphnia. The plankton data were collected at Greifensee using a novel dual-magnification dark-field imaging microscope deployed at 3 meters deep in hourly temporal frequency (Merz et al. 2021). The high temporal resolution of the dataset captures processes that occur at short time-scales, including diel vertical migrations of zooplankton (see Fig. 8 in Merz et al. 2021). Although our model does not resolve the vertical distribution of plankton in the water column, our results are constrained, through model parameterization, by time-integrated community properties characterizing mean zooplankton size and grazing dynamics in the well-mixed epilimnion of Greifensee.

The measured abundance data were in units of ROI/sec (Regions Of Interest per second). This pseudo-unit is a statistical output derived from each detected and processed object by the imaging microscope and is analogous to object counts for both unicellular and colonial phytoplankton and zooplankton individuals (Merz et al. 2021). We converted these data into plankton abundances (individual L^{-1}) based on laboratory-calibrated functions (Fig. S2, Table S2), derived by Merz et al. (2021). The plankton abundances were then converted into phosphorus-equivalent biomass concentrations, following previously reported scaling functions based on the size of phytoplankton and zooplankton (section S4 in Supporting Information).

As key physiological and ecological traits – such as nutrient uptake, growth rate, and grazing vulnerability – scale predictably with cell size (Litchman and Klausmeier, 2008; Edwards et al. 2012), we modeled and analyzed the plankton community based on cell size rather than taxonomy. Size-based approaches provide a unified framework for representing functional diversity, as most major phytoplankton groups can be characterized along a continuous size spectrum (Litchman and Klausmeier 2008). The size data were collected as bio-area (μm^2) and converted to biovolume based on an empirical scaling relationship (Table S2). In our model, cell size is represented in terms of equivalent spherical diameter, ESD (μm), which accounts for differences in size and geometric shape of plankton (Jennings and Paslow 1988). Therefore, the phytoplankton community, represented in the model by 10 logarithmically spaced size classes spanning across a size spectrum from 1 to 200 μm ESD (i.e., equivalent to biovolume of 5×10^{-1} to 4×10^{6} μm^3), encompasses the full range of taxa observed at Greifensee, including the six most dominant groups used for calibration (Table S5). For

zooplankton, we considered two size classes, Z_1 and Z_2, of 20 and 600 μm, respectively. We started our simulations with an initial concentration of 6.25×10^{-4} μM P for all planktonic organisms. We analyzed the phytoplankton size composition in macroecological terms (e.g., Acevedo-Trejos et al. 2018; To et al. 2024), by considering a community-weighted mean cell biovolume, $\overline{V^A}$:

$$\overline{V^A} = \frac{\sum_i^n A_i \, V_i^A}{\sum_i^n A_i} \tag{S16}$$

where n denotes the total number of size classes, and A_i and V_i^A denote, respectively, total biomass and biovolume of the ith class of phytoplankton.

All model parameters were obtained from literature (Table S1). In order to find the best fit between model results and observations, parameters regulating nutrient uptake and light harvesting by phytoplankton, as well as zooplankton grazing (α_{P_I}, K_P, $\alpha_{\psi_{max}}$, $\alpha_{A_{opt}}$, $\theta_j^{Z_1} and\ \theta_j^{Z_2}$), were fine-tuned manually within literature-reported ranges (Fig. S6, Table S1) and the best fit was determined based on visual inspection. This calibration produced a model run generating results that best matched the observations, hereafter referred to as the 'standard model run'. Nutrient concentrations and plankton abundances observed at Greifensee over three consecutive years were aggregated into a single year to produce a representative annual cycle for every variable (section S4 in Supporting Information). Given our focus on the mean annual state of the lake, the model was run to steady-state over 10 years using identical annual forcing, and only the final year was used for analysis. Interannual variability was not explored, as it was beyond the scope of our study. Additionally, we performed a sensitivity analysis to evaluate the effects of uncertainties related to those parameters (n=14) that were not tuned (see section S5 in Supporting Information).

S4. Conversions of observations and model results

We used two raw plankton size datasets that are obtained from a field-deployed imaging device dated from March 2019 to December 2022 (available in Supplementary Data, doi:10.5281/zenodo.11390345). The plankton data for Greifensee, collected by the dual-magnification dark-field imaging microscope (Orenstein et al. 2020), contain plankton abundance in terms of ROI/second (Regions Of Interest per second) and plankton size estimates in terms of bio-area (mm^2), minimum axis length (mm), and maximum axis length (mm). ROI/sec is a pseudo-unit derived from the statistical output of each detected and processed object by the imaging microscope that converts ROIs into image mosaics. In general, this unit is analogous to object counts for both unicellular and colonial phytoplankton and zooplankton individuals (Merz et al. 2021). To be comparable with model outputs, these data were converted into abundances (individual L^{-1}) based on laboratory calibration functions (Merz et al. 2021). After that, the abundances were converted into community biomass concentration (µM P), based on the size data of phytoplankton and zooplankton. These conversions are explained in the following and detailed equations are given in Table S1.

Plankton abundances

To obtain plankton abundances, we adopted the two calibrated functions derived by Merz et al. (2021) and presented in Fig. S2. One equation was derived for the magnifier 5p0x (for smaller objects, between 10 and 150 µm), whereas another was for the magnifier 0p5x (for larger objects, between 100 µm and 1 cm). We applied the equations to the plankton data according to the magnifier that produced that data (Table S1). The phytoplankton groups identified by the 5p0x magnifier are *centric diatom*, *cryptophyceae*, *rhodomonas*, and *chlorophytes*, whereas those identified by the 0p5x magnifier are *dinobryon* and *uroglena*. The zooplankton groups captured from 5p0x and 0p5x are, respectively, *ciliates* and *daphnia*.

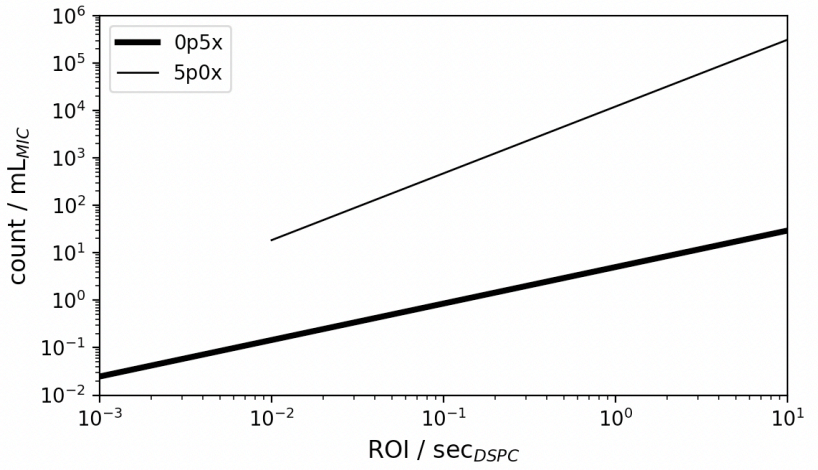


Fig. S2. Laboratory calibration of density estimated by microscopic count (count/mL) against density estimated by the plankton camera (ROI/sec). The functions are replotted from Merz et al. (2021): 5p0x (r^2 = 0.889) and 0p5x (r^2 = 0.726), Table S2.

Phosphorus-equivalent phytoplankton biovolume

We transformed the bio-area (mm^2) estimates of phytoplankton to biovolume based on the relationship established by Ryabov et al. (2021), followed by calculating the phosphorus content using an allometric scaling function (Fig. S3) established by Hillebrand et al. (2022). The equations are shown in Table S2.

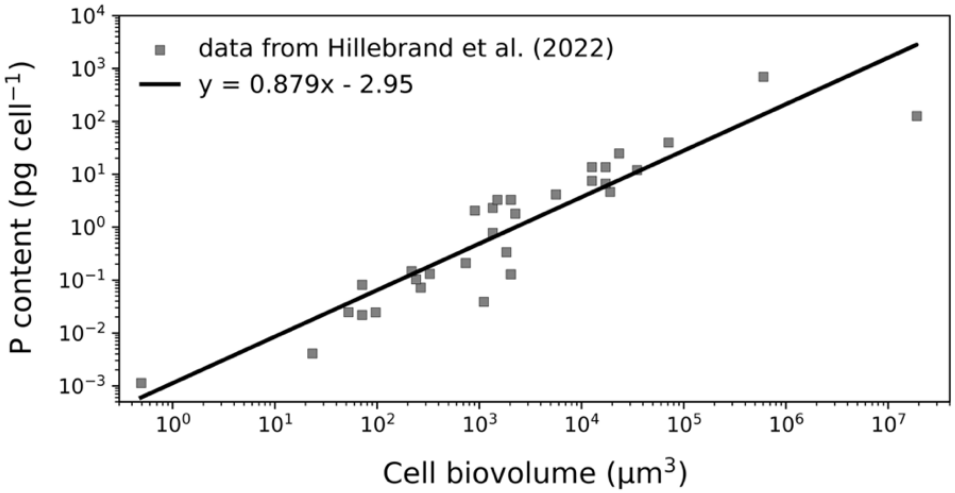


Fig. S3. The scaling function of phosphate content per cell in relation to phytoplankton biovolume (μm^3). The data (grey squares) are from Hillebrand et al. (2022) and the function (solid dark line) is a linear regression on the original data (r^2 = 0.924).

Phosphorus-equivalent zooplankton biovolume

Ciliates and Daphnia: The biovolume of ciliates was approximated assuming that the general shape of ciliates is a standard ellipsoid, following Hillebrand et al. (1999). We use the major and minor axis length data of the ciliates to represent the principal diameters of the ellipsoidal ciliate and assume that the third dimension of the ellipsoidal ciliate is equal to the minor axis length (Table S1). We obtain the carbon content per ciliate individual in relation to size according to the size-scaling relationship of Putt and Stoecker (1989) and the stoichiometric transformation of carbon to phosphorus is based on a range within 80-120 measured for a freshwater ciliates *Euplotes sp.* (Golz et al. 2015).We used the major axis length data of daphnia to estimate their carbon content based on a body length to weight relationship in Schalau et al. (2008) and considered a correction factor of 0.75 from the major axis length to body length. The stoichiometric conversion of carbon to phosphorus is based on a ranges from 80 to 385 in previous modelling studies (reviewed in Khattak et al. 2018), Table S1. The detailed conversion is shown in Table S1, and the values of the conversion coefficients are given in Table S2. Those values were manually optimized by comparing and matching the plankton abundances obtained from the observations with the abundances produced by the model.

S5. Sensitivity analysis on standard model parameters

We conducted a sensitivity analysis to evaluate the uncertainty of those parameters that could not be directly estimated from Greifensee data (Table S1). A total of 14 parameters were tested, except the thermal traits that were tested in a different sensitivity analysis (Fig. S5) and the parameters that were already calibrated by matching model results to observations (Fig. S6). We varied the 14 parameters by ±25% from standard value and quantified the percentage change on the annual median for four model variables, nutrient concentrations, community-integrated phytoplankton and zooplankton biomass, and community-mean phytoplankton biovolume (Table S6). This sensitivity test is simulated for the standard model under the default physical forcing.

The result shows that nutrient concentration and phytoplankton biomass are generally sensitive to the parameter tested (Fig. S4A, B). Zooplankton biomass and community-mean phytoplankton biovolume are generally not very sensitive to the parameter tested (Fig. S4C, D). Nutrient concentrations are most sensitive (affect nutrient values by max. ±60%) to the slope and intercept of allometric relationships for phytoplankton growth ($\alpha_{\mu_{max}}$, $\beta_{\mu_{max}}$), zooplankton grazing ($\beta_{\psi_{max}}$), and optimal phytoplankton size for zooplankton ($\beta_{A_{opt}}$), Fig. S4A. This suggests that nutrient concentration is sensitive to growth and grazing parameters that vary with size-dependencies. Phytoplankton concentrations are more sensitive (altered by max. ±60%) to parameters that control grazing, including zooplankton mortality (ϕ_Z), fraction of grazed material that is not ingested, i.e., sloppy feeding (ε), fraction of grazed material that is assimilated into zooplankton growth (γ), half-saturation constant for grazing (K_A), and the intercept of allometric relationship of optimal phytoplankton size for zooplankton grazing ($\beta_{A_{opt}}$), Fig. S4B. Other parameters produced weaker effects to the annual median value of nutrient and phytoplankton biomass, i.e., within ±30% (Table S6).

Our result shows moderate changes in zooplankton (within about ±15%), Table S6. The community-mean phytoplankton biovolume is insensitive to most parameters (within ±10%), except to the +25% slope of allometric relationships for maximum phytoplankton growth ($\alpha_{\mu_{max}}$) and the +25% intercept of allometric relationships of optimal phytoplankton size for zooplankton grazing ($\beta_{A_{opt}}$), Fig. S4D. Increasing these two parameters favors small phytoplankton by decreasing the growth ability of larger phytoplankton and by weakening grazing on small phytoplankton (grazing shifts to target larger phytoplankton). Both parameters produce a strong decline in mean phytoplankton biovolume (Fig. S4D).

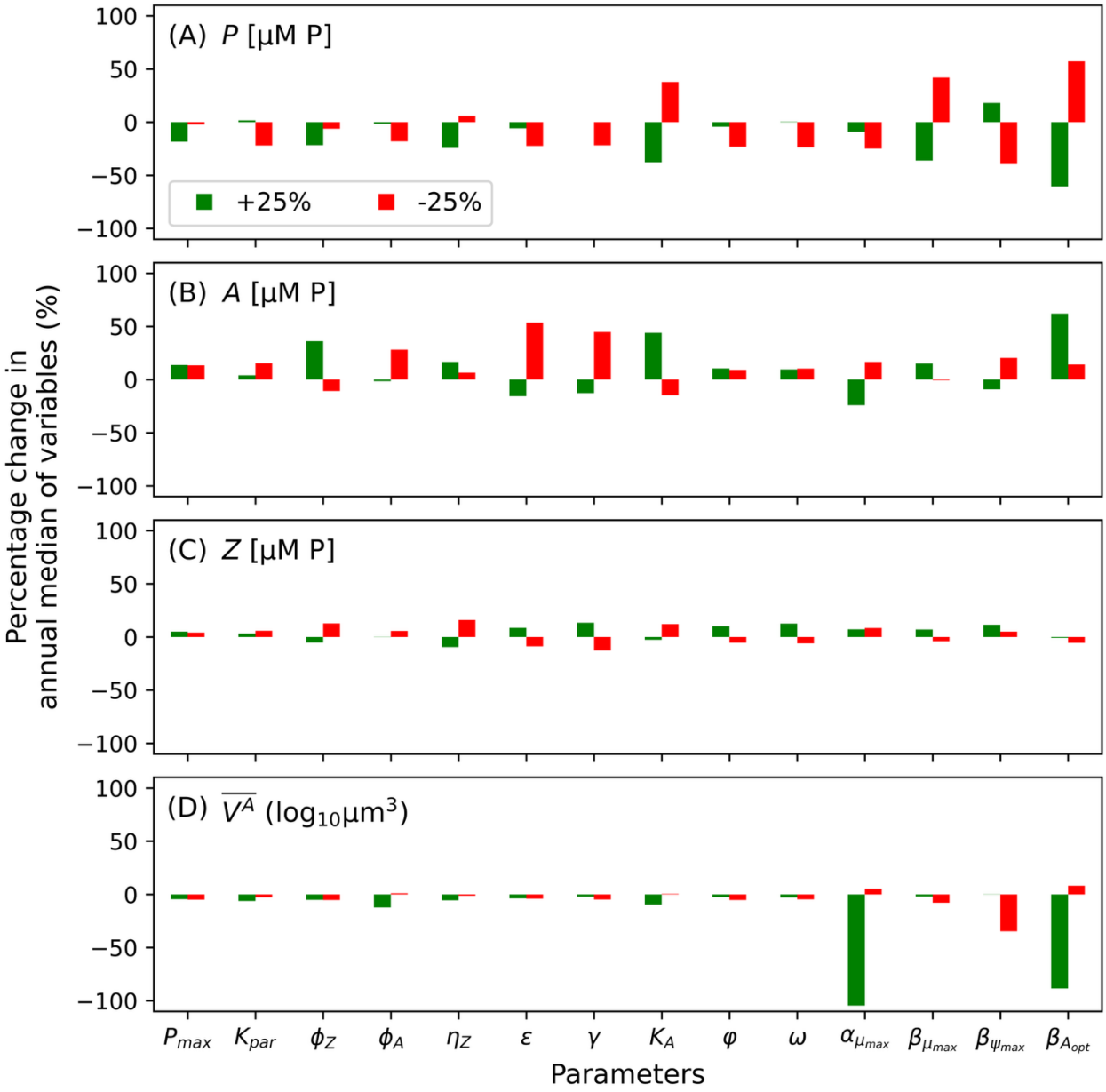


Fig. S4. Results for the sensitivity analysis on fourteen model parameters. These are parameters that were neither examined in sensitivity analyses nor calibrated with observations. The result shows changes in annual median of four model variables, (A) nutrient concentration in the epilimnion, (B) phytoplankton biomass, (C) zooplankton biomass, and (D) phytoplankton mean biovolume, by +25% (green bars) and -25% (red bars) defaulted parameter values (Table S1). The percentage change values are reported in Table S6.

S6. Sensitivity test on alterations of lake stratification

In a warming world, higher temperature and longer stratification periods are factors (Woolway et al. 2020, 2021) that can affect nutrient transport (Behrenfeld et al. 2006), plankton biomass, and plankton phenology (Winder and Sommer 2012). Earlier onsets of stratification and longer periods of stratification are expected to occur in multiple temperate lakes across the Northern Hemisphere (Winder and Schindler 2004; Shatwell et al. 2019; Woolway et al. 2021).

Since our ecological projection did not consider any changes in the lake stratification, we performed a sensitivity test varying the lake mixing dynamics under the most severe climate change scenario, RCP 8.5. Lake stratification in our model is governed by two features, variations of the mixed layer depth (MLD) in combination with a cross-thermocline diffusive mixing, captured by a constant parameter (ω). Based on the estimation of the median change in stratification period by Woolway et al. (2021), we created a prolonged lake stratification that starts 22 days earlier and ends 11 days later (Fig. S5A). This leads to a longer summer stratification, by 33 days. In another case, we reduced the diffusive mixing coefficient by 50%. This parameter determines the strength of diffusive mixing between hypolimnion and epilimnion.

The model output shows that, these variations produce effects to all variables. The more prominent effects include a lower level of nutrient concentration throughout the year (Fig. S5B) and an earlier zooplankton spring peak under a prolonged stratification (Fig. S5D). A weaker diffusive mixing produces a lower level of nutrient accumulation (30% reduction in annual median) in summer compared to the standard mixing dynamics (Fig. S5B, inset). This is likely due to the reduced amount of nutrient recharged from the hypolimnion. Additionally, in our model, an earlier lake stratification leads to earlier development of zooplankton peak in spring. That is because the zooplankton peak is sensitive to the start of stratification (since the shallowing of mixed layer depth increases biomass concentration, Eq. S2), but they are not affected by changes in diffusive mixing. The alteration in lake stratification produces only minor changes to the biomass dynamics and community mean biovolume of phytoplankton (Fig. S5C, E).

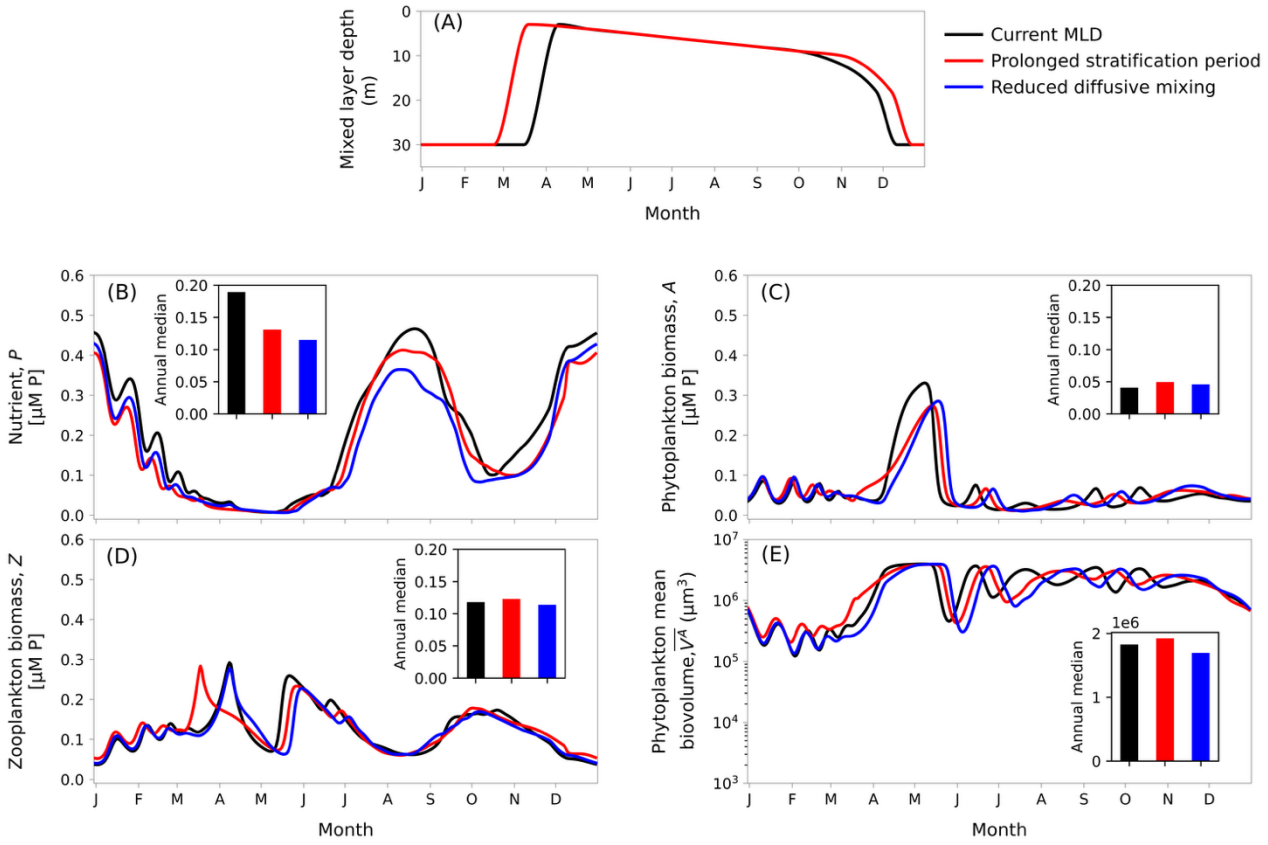


Fig. S5. Results of sensitivity test on alterations of lake stratification compared to the standard mixing regime under the most severe climate change scenario (RCP 8.5, water temperature increases by 4.2 °C throughout the year). (A) Standard mixed layer depth (black), future mixed layer depth produced by (1) a prolonged stratification period (red) and (2) a weakened diffusive mixing between hypolimnion and epilimnion (blue). The results show, for the three mixing regimes, temporal dynamics of (B) nutrient concentration in the epilimnion, (C) phytoplankton biomass, (D) zooplankton biomass, and (E) phytoplankton mean biovolume. The insets show the annual median value for different alterations. The model is run to a steady annual cycle over a period of 10 years with repeated annual forcing, the results shown are those of the 10th year.

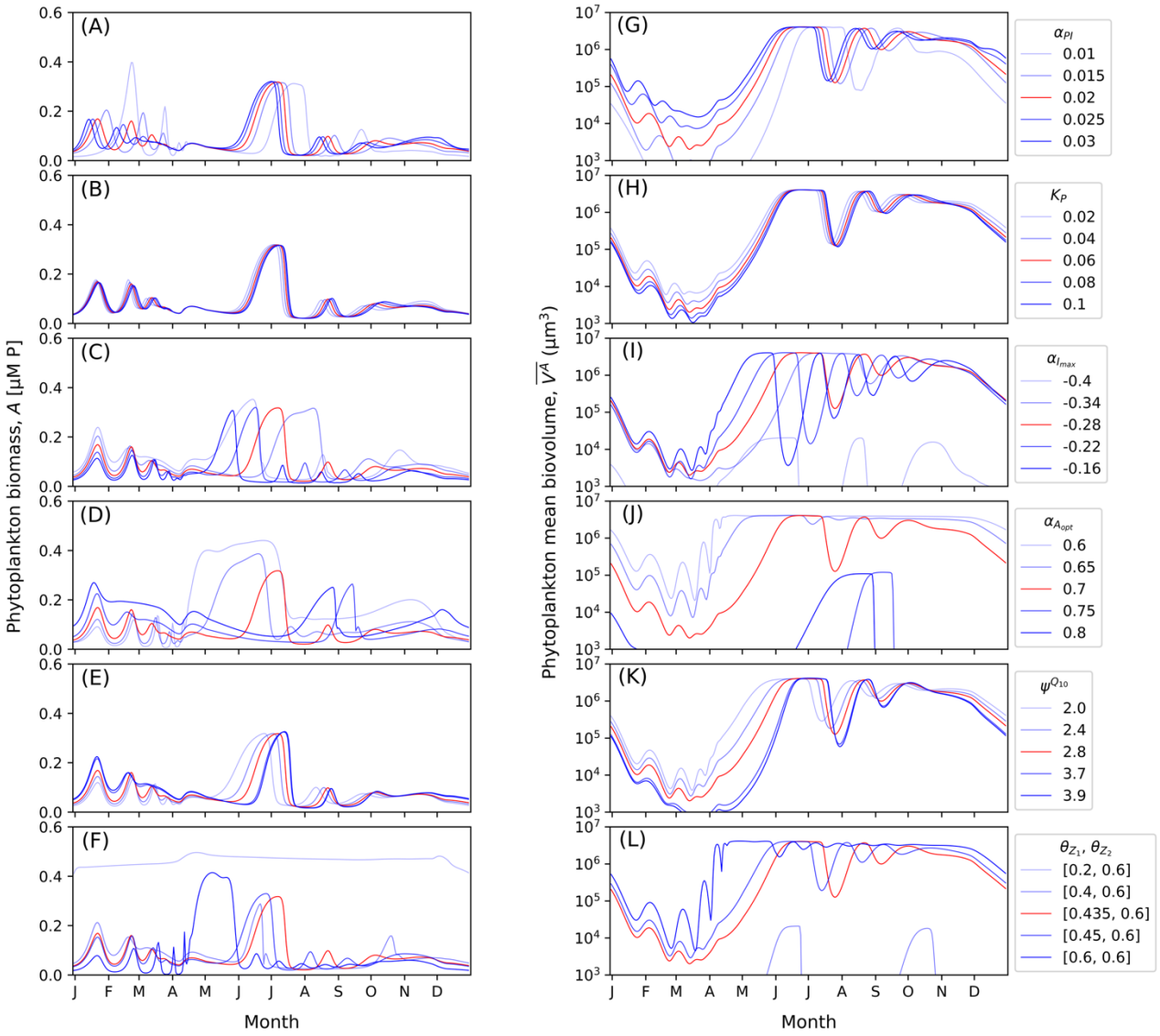
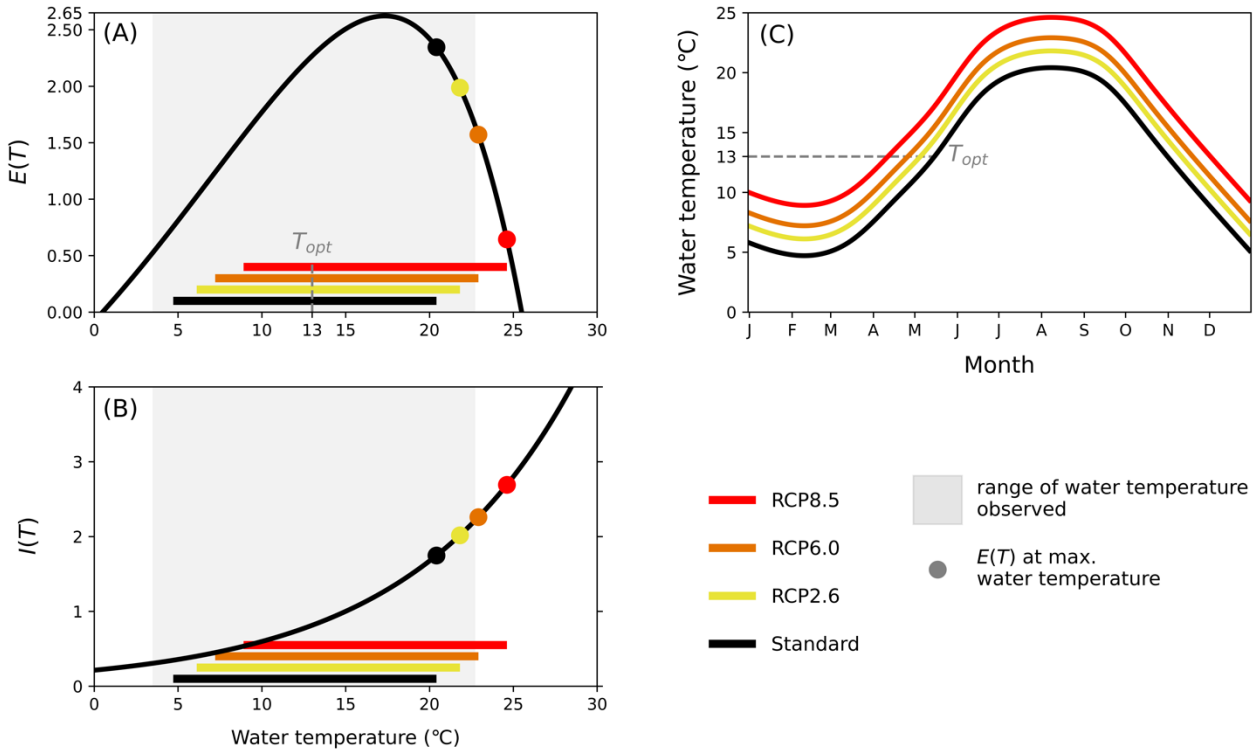


Fig. S6. Selected results for the calibrated parameters. (A, G) initial slope of P-I curve, α_{P_I}; (B, H) half saturation constant for nutrient, K_P; (C, I) slope of the allometric relationship for maximum ingestion rate, $\alpha_{I_{max}}$; (D, J) slope of the allometric relationship for optimal phytoplankton size, $\alpha_{A_{opt}}$; (E, K) Q_{10} coefficient for maximum ingestion rate, $\psi^{Q_{10}}$; and (F, L) size-grazing breadth for Z_1 and Z_2, $\theta_j^{Z_1}$ & $\theta_j^{Z_2}$. The results for (A-E) phytoplankton biomass concentrations, A, and (F-J) phytoplankton community mean biovolume, $\overline{V^A}$. The red line indicates the value used for the standard model run.



406

407 **Fig. S7**. Temperature dependence curves for (A) phytoplankton growth, $E(T)$, and (B) zooplankton
408 grazing, $I(T)$, for different scenarios: standard (dark), RCP 2.6 (yellow), RCP 6.0 (orange), and RCP
409 8.5 (red). (C) Optimal temperature for growth, T_{opt}. The standard range of water temperature is
410 represented by the black line. The climate change scenarios simulate an increase in the standard range
411 of water temperature by, respectively, 1.4 °C (RCP 2.6, yellow bar), 2.5 °C (RCP 6.0, orange bar),
412 and 4.2 °C (RCP 8.5, red bar). The circles (A, B) indicate the level of temperature dependence at the
413 hottest day of the year from different scenarios and show that phytoplankton growth is becoming
414 increasingly suboptimal while zooplankton grazing intensifies.

415

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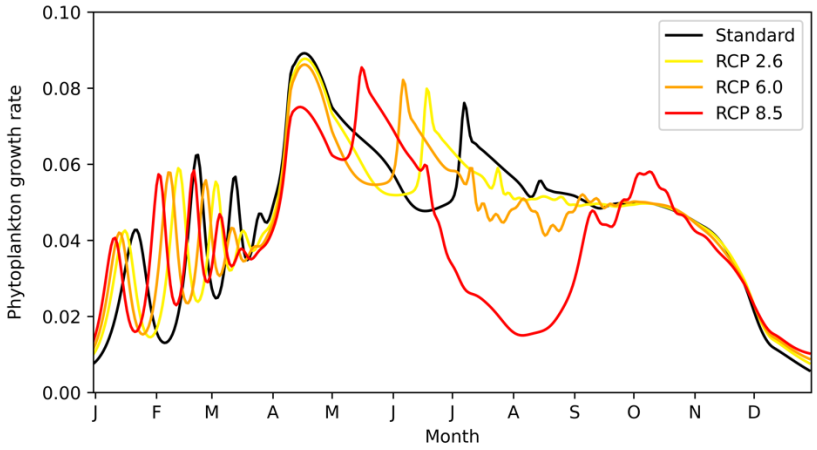


Fig. S8. Phytoplankton growth rates, $\sum \mu_i$ (Eq. S8), in different scenarios: standard (dark), RCP 2.6 (yellow), RCP 6.0 (orange), and RCP 8.5 (red). It shows a pronounced decline in summer phytoplankton growth under RCP 8.5.

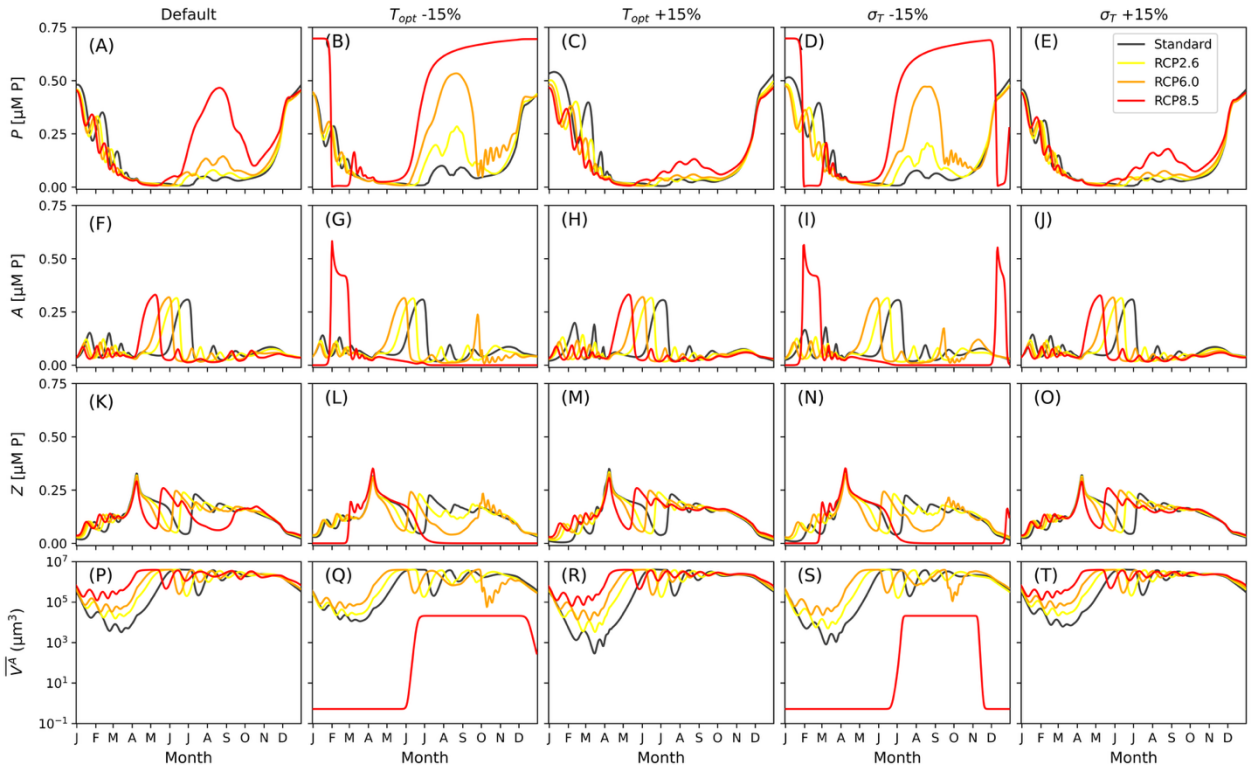


Fig. S9. Results of the sensitivity analysis (±15%) related to the two thermal traits, T_{opt} and σ_T. The result shows the temporal dynamics of four model variables: nutrient concentration, P (A-E); phytoplankton biomass concentration, A (F-J); zooplankton biomass concentration, Z (K-O); and phytoplankton community mean biovolume, $\overline{V^A}$ (P-T), under the different scenarios: standard (dark), RCP 2.6 (yellow), RCP 6.0 (orange), and RCP 8.5 (red).

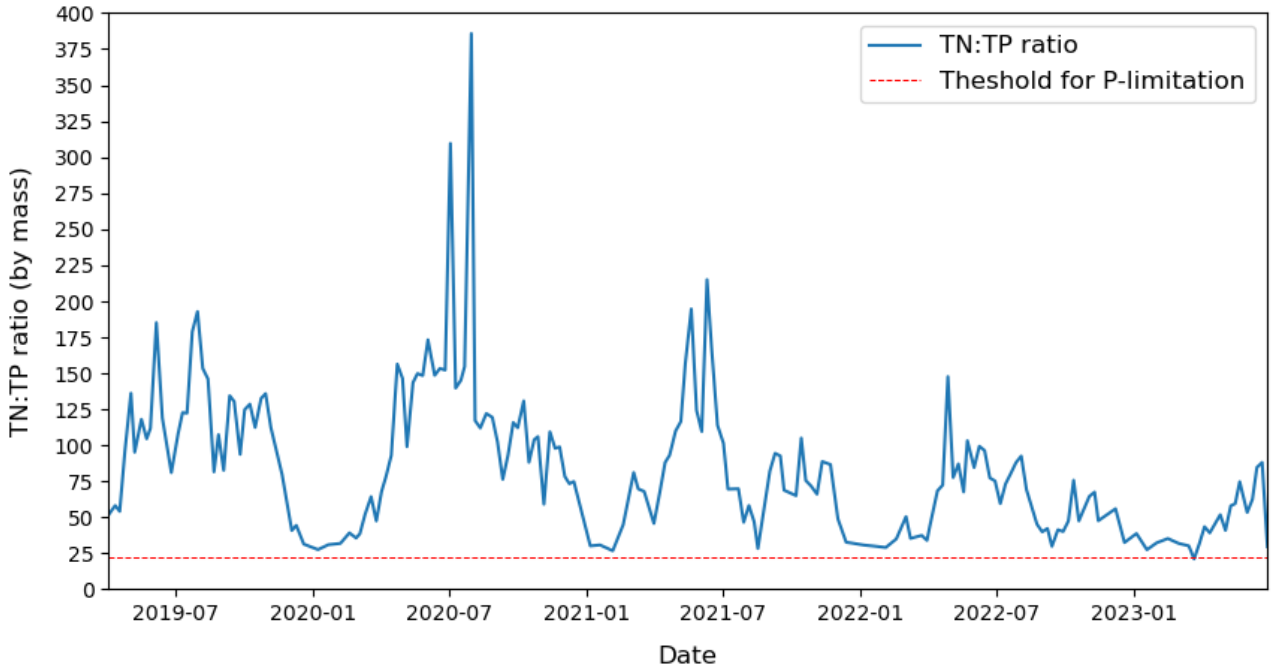


Fig. S10. Temporal trend of total nitrogen to total phosphorus (TN:TP) ratio at Greifensee at 3 m depth, during the period 2019 – 2023 (data from Eyring et al. 2025). The red dashed line marks the threshold (22.58) above which water bodies are characterized as P-limited, according to Guildford and Hecky (2000).

Table S1. List of model parameters with corresponding symbols, standard values, parameter range, units, and literature sources. The asterisk (*) indicates parameter values that were fine-tuned within literature ranges to produce a 'standard model run' (i.e., a model run that generates results best fitting observations).

Parameter	Symbols	Standard Value	Parameter range	Units	Sources for parameter/ parameter range
Initial slope of P-I curve	α_{P_I}	0.02*	0.02 - 0.1	Ein m^{-2} d^{-1}	Anderson et al. (2015)
Maximum photosynthesis rate	P_{max}	1.1	-	d^{-1}	To et al. (2024)
Light attenuation coefficient	K_{par}	0.1	0.08 – 0.5	m^{-1}	Chen et al. (2002); Genkai-Kato et al. (2012)
Thermal tolerance	σ_T	12.5	-	℃	Greifensee data
Thermal optima	T_{opt}	13	-	℃	Greifensee data
Half-saturation constant for nutrient (phosphorus)	K_P	0.06*	0.06 -0.1	µM P	Edwards et al. (2012)
Slope of the allometric relationship for maximum growth rate, $\mu_{max_i}(S_i^A)$	$\alpha_{\mu_{max}}$	-0.36	-	-	Edwards et al. (2012)
Intercept of the allometric relationship for maximum growth rate, $\mu_{max_i}(S_i^A)$	$\beta_{\mu_{max}}$	$10^{0.69}$	-	d^{-1}	Edwards et al. (2012)
Cross-thermocline diffusive mixing coefficient	ω	0.1	-	m day^{-1}	Fasham et al. (1990)
Natural mortality - Phytoplankton	ϕ_A	0.1	-	d^{-1}	this study
Natural mortality – Zooplankton	ϕ_Z	0.1	-	d^{-1}	To et al. (2024)
Higher-order mortality rate	η_Z	0.34	-	d^{-1}	Oschlies & Schartau (2005)
Fraction of grazing not egested	ε	0.69	-	-	Fasham et al. (1990)

Parameter	Symbols	Standard Value	Parameter range	Units	Sources for parameter/ parameter range
Fraction of grazing assimilated to growth	γ	0.75	-	-	Oschlies & Schartau (2005)
Remineralization rate	φ	0.15	-	d^{-1}	Kerimoglu et al. (2017)
Half-saturation constant to phytoplankton cells	K_A	0.19	-	µM P	Kerimoglu et al. (2017)
Size-grazing breadth - Z_1, Z_2	${\theta_j}^{Z_1}$, ${\theta_j}^{Z_2}$	0.435, 0.6*	0.2-0.8	µm ESD	Banas (2011)
Slope of the allometric relationship for maximum ingestion rate, ${\psi_{max}}_j(S_j^Z)$	$\alpha_{\psi_{max}}$	-0.28*	-0.25 - -0.4	-	Hansen et al. (1997)
Intercept of the allometric relationship for the maximum ingestion rate, ${\psi_{max}}_j(S_j^Z)$	$\beta_{\psi_{max}}$	23	22-26	d^{-1}	Hansen et al. (1997)
Coefficient for zooplankton grazing	Q_{10}	2.8*	2.0 - 3.9	-	Data in Hansen et al. (1997)
Reference temperature for zooplankton grazing	T_{ref}	15	-	°C	Hansen et al. (1997); Kerimoglu et al. (2017)
Slope of the allometric relationship for optima phytoplankton size, ${A_{opt}}_j(S_j^Z)$	$\alpha_{A_{opt}}$	0.7*	-	-	Hansen et al. (1997)
Intercept of the allometric relationship for optimal phytoplankton size, ${A_{opt}}_j(S_j^Z)$	$\beta_{A_{opt}}$	0.45		µm ESD	Hansen et al. (1997)
Cell sizes – phytoplankton	S_i^A	1-200	-	µm ESD	Greifensee data
Body sizes – zooplankton	S_j^Z	20 (Z_1) 600 (Z_2)	-	µm ESD	Greifensee data

Parameter	Symbols	Standard Value	Parameter range	Units	Sources for parameter/ parameter range
Average inorganic phosphate released from the hypolimnion and initial condition for phosphate concentration	P_0	0.75	-	µM P	Greifensee data
Initial conditions for phytoplankton, zooplankton, and detritus	A_0, Z_0, D_0	6.25e-4	-	µM P	this study

Table S2. Transformation of model and data output with physical, stoichiometric, and size-scaling coefficient. The references for the equations are provided below the equations and those for specific coefficient are given in Table S3

		Solar irradiance ($W\ m^{-2}$)	Abundance data		Phytoplankton size		Ciliates Minimum and maximum axis length (mm)	Ciliates carbon content ($pg\ C\ ind^{-1}$)	Daphnia carbon content ($\mu g\ C\ ind^{-1}$)
			ROI/sec (<100 µm)	ROI/sec (>100 µm)	Bio-area, A (μm^2)	Modeled ESD (µm)			
Physical	Net Short-wave solar irradiance i.e. light usable for photosynthesis ($W\ m^{-2}$)	0.45 (Table S3)							
	Photosynthetically Active radiance ($E\ m^{-2}\ d^{-1}$)	$\frac{5 \cdot 86400}{10^6}$ (Table S3)							
Abundance	Phytoplankton ($ind\ L^{-1}$)		$10^{\frac{2.9}{0.71}} \cdot \left(10^{\mathrm{ROI/sec}}\right)^{\frac{1}{0.71}}$ Merz et al. (2021)						
	Zooplankton ($ind\ L^{-1}$)		$10^{\frac{0.91}{1.3}} \cdot \left(10^{\mathrm{ROI/sec}}\right)^{\frac{1}{1.3}}$ Merz et al. (2021)						
Phytoplankton	Biovolume, V (μm^3)				$\left(\frac{A}{\omega}\right)^{\frac{1}{0.68}}$ Ryabov et al. (2021)	$\frac{4}{3} \cdot \pi \cdot \left(\frac{ESD}{2}\right)^3$ Hillebrand et al. (1999)			
Zooplankton	Ciliates biovolume, V (μm^2)						$\frac{\pi}{6} \cdot a \cdot b \cdot h \cdot 10^3$ Hillebrand et al. (1999)	$10^{0.12} \cdot V^{0.19}$ Putt and Stoecker (1989)	
	Daphnia maximum axis length, L (mm)								$1.6 \cdot L^3$ Schalau et al. (2008)
Stoichiometry	Phosphate content per individual cell ($pg\ P\ ind^{-1}$)				$10^{-2.95} \cdot V^{0.88}$ (Hillebrand et al., 2022)				
	Phosphorus-equivalent biomass concentration ($\mu mol\ P\ ind^{-1}$)				$\frac{1}{\mathrm{mw\ P}} \cdot 10^{-6}$ (Table S3)			$\frac{1}{\mathrm{mw\ C}} \cdot \frac{1}{\mathrm{C{:}\ P_{Ci}}} \cdot 10^{-6}$ (Table S3)	$\frac{1}{\mathrm{mw\ C}} \cdot \frac{1}{\mathrm{C{:}\ P_{Da}}}$ (Table S3)

Table S3. Parameters for physical and stoichiometric conversion and their descriptions, standard values, parameter range, units, and sources for the parameter or parameter range. Those that are provided with a range was calibrated by manually comparing the range of biomass values obtained from the data with those from the model.

Parameter	Description	Standard Value	Parameter range	Units	Source for parameter
-	Solar irradiance to net short-wave solar irradiance	0.45	0.41-0.5	W m^{-2}	Howell et al. (1983); Tsubo and Walker (2005)
-	Conversion of energy flow density to photon flux density	5	4-5	µE m^{-2} s^{-1} $(mol\ P)^{-1}$	Lampert and Sommer (2007)
ω	Offset value in data of Ryabov et al. (2021)	$10^{0.25}$	-	μm^2	Calibrated in this study
mw P	Molar weight of phosphorus	31.97	-	µg P	Standard unit
mw C	Molar weight of carbon	12.01	-	µg C	Standard unit
C: P_{Ci}	C:P ratio for ciliates	100	80-120	mol C $(mol\ P)^{-1}$	Golz et al. (2015)
C: P_{Da}	C:P ratio for daphnia	350	80-385	mol C $(mol\ P)^{-1}$	Khattak et al. (2018)

Table S4. Information for the increase in lake water surface temperature for future projections (i.e. year 2099 or 2100) and the corresponding sources.

RCP scenarios	Value used for projections	Range of changes in lake water surface temperature (°C)	Location	References
RCP 2.6	+1.4°C	1.36	Lake Erken, Sweden	Ayala et al. (2020)[1]
(low-emission scenario)		1.4	10 Polish lakes	Czernecki and Ptak (2018)[2]
		0.09 decade^{-1}	Rappbode Reservoir, Germany	Mi et al. (2020)[3]
		0.51 century^{-1}	Lake Kozjak, Croatia	Šarović and Klaić (2023)[4]
RCP 6.0	+2.5°C	2.31	Lake Erken, Sweden	Ayala et al. (2020)
(medium-high-emission scenario)		1.8 – 2.7	The Great Lake region	Xue et al. (2022)[5]
		0.32 decade^{-1}	Rappbode Reservoir, Germany	Mi et al. (2020)
		1.41 century^{-1}	Lake Kozjak, Croatia	Šarović and Klaić (2023)
RCP 8.5	+4.2°C	4.2	10 Polish lakes	Czernecki and Ptak (2018)
(high-emission scenario)		4.1 – 5.0	The Great Lake region	Xue et al. (2022)
		0.5 decade^{-1}	Rappbode Reservoir, Germany	Mi et al. (2020)
		4.51 century^{-1}	Lake Kozjak, Croatia	Šarović and Klaić (2023)

[1] Ayala et al. (2020) adopts the one-dimensional hydrodynamic model, GOTM, to simulate the lake surface water temperature in future scenarios that are projected by the GCMs from ISIMIP scheme. Their long-term simulations were forced with daily or hourly meteorological data.
[2] Czernecki and Ptak (2017) simulates future lake surface water temperature using empirical-statistical downscaling modelling techniques, based on simulations of 33 atmosphere-ocean coupled GCMs obtained from the CMIP5 project.
[3] Mi et al. (2020) uses a well-developed hydrodynamic model and consider a set of reanalyzed daily meteorological input (bias-corrected for ISIMIP).
[4] Šarović et al. (2023) uses a one-dimension energy budget model to simulate changes in mean lake water temperature, based on projected meteorological data (i.e. CORDEX) taken from the World Climate Research Programme.
[5] Xue et al. (2022) couples a regional climate model with a three-dimensional hydrodynamic lake model, GLARM, using GCMs obtained from the CMIP5 project.

Table S5. Size distributions plankton groups observed at Greifensee and selected for our study. The bio-area values were converted to Biovolume (μm^3), based on an empirical scaling relationship (Table S2).

	Mean bio-area (μm^2)	Mean Biovolume (μm^3)	Mean Equivalent Spherical Diameter (μm)
Phytoplankton			
Uroglena	36926.5	$2.5x10^6$	156.1
Dinobryon	2453.9	$5.0x10^4$	40.6
Centric diatom	803.1	$8.3x10^3$	24.5
Cryptophytes	438.6	$3.3x10^3$	18.4
Chlorophytes	224.0	$1.3x10^3$	13.1
Rhodomonas	151.5	$6.9x10^2$	11.0
Zooplankton			
Daphnia	342510.7	$6.0x10^7$	479.2
Ciliates	933.2	$1.0x10^4$	26.4

Table S6. Percentage change on the annual median value of four model variables: nutrient concentration in the epilimnion, P, phytoplankton biomass, A, zooplankton biomass, Z, and phytoplankton mean biovolume, $\overline{V^A}$, under ±25% change in fourteen model parameters for the standard model run.

Sensitivity of variables	P [µM P]		A [µM P]		Z [µM P]		$\overline{V^A}$ ($\log_{10}$ µm^3)	
% Change in parameter values	**+25%**	**-25%**	**+25%**	**-25%**	**+25%**	**-25%**	**+25%**	**-25%**
P_{max}	-18.31	-2.31	13.68	13.38	5.06	4.1	-4.48	-4.93
K_{par}	1.74	-21.85	4.05	15.44	3.15	5.81	-6.11	-2.85
ϕ_A	-21.68	-6.32	36.17	-10.73	-5.16	12.7	-5.12	-5.25
ϕ_Z	-1.53	-18.06	-1.45	28.14	0.18	5.61	-12.33	0.97
η_Z	-24.25	5.87	16.54	6.52	-9.53	15.89	-5.49	-1.28
ε	-5.83	-22.43	-15.71	53.66	8.6	-8.74	-3.68	-3.96
γ	0.11	-21.7	-12.7	44.72	13.35	-12.78	-1.98	-4.79
K_A	-37.68	37.81	43.86	-14.7	-2.56	12.19	-9.61	0.37
φ	-4.31	-23.1	10.41	9.03	10.06	-5.42	-2.6	-5.31
ω	0.35	-23.68	9.53	10.33	12.62	-5.99	-2.95	-4.62
$\alpha_{\mu_{max}}$	-9.03	-24.79	-24.05	16.65	7.25	8.42	-104.65	5.13
$\beta_{\mu_{max}}$	-36.06	42.03	15.11	-0.66	6.99	-4.0	-1.87	-7.88
$\beta_{\psi_{max}}$	18.1	-39.41	-9.16	20.39	11.55	5.1	0.09	-34.71
$\beta_{A_{opt}}$	-60.51	57.14	61.83	14.11	-0.77	-5.48	-88.48	8.12

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